NOTE: ALL CIVIL ENGINEERING CONSTRUCTION WORKS TO BE CARRIED OUT IN ACCORDANCE WITH AUTHORITY (INCLUDING LGA AND WATER AUTHORITY) STANDARDS AND GUIDELINES.THE AFOREMENTIONED GUIDELINES INCLUSIVE OF ALL SPECIFICATIONS TAKE PRECEDENCE OVER NOTES PROVIDED BELOW.

SITEWORKS

1. ALL WORKS TO BE IN ACCORDANCE WITH RELEVANT LOCAL COUNCIL / REGULATORY AUTHORITIES REQUIREMENTS, ALL SPECIFICATIONS AND AUSTRALIAN STANDARDS. CONFLICTS BETWEEN SAID DOCUMENTS SHALL BE REFERRED TO THE SUPERINTENDENT FOR DIRECTION.
2. THE CONTRACTOR IS TO DESIGN, OBTAIN APPROVALS AND CARRY OUT REQUIRED TEMPORARY TRAFFIC CONTROL PROCEDURES DURING CONSTRUCTION IN ACCORDANCE WITH ALL REGULATORY AUTHORITIES, INCLUSIVE OF LOCAL COUNCIL REGULATIONS AND REQUIREMENTS.
3. THE CONTRACTOR IS TO OBTAIN ALL AUTHORITY APPROVALS AS REQUIRED PRIOR TO COMMENCEMENT OF WORKS.
4. RESTORE ALL PAVED, COVERED, GRASSED AND LANDSCAPED AREAS TO THEIR ORIGINAL CONDITION OR AS DIRECTED BY THE SITE SUPERINTENDENT ON COMPLETION OF WORKS. WHERE PLANTING OF NEW GRASS IS NECESSARY REFER TO LANDSCAPE ARCHITECT AND / OR ARCHITECT DOCUMENTATION.
5. ON COMPLETION OF ANY TRENCHING WORKS, ALL DISTURBED AREAS SHALL BE RESTORED TO THEIR ORIGINAL CONDITION OR AS DIRECTED BY THE SITE SUPERINTENDENT, INCLUDING KERBS, FOOTPATHS, CONCRETE AREAS, GRAVEL, GRASSED AREAS AND ROAD PAVEMENTS.
6. THE CONTRACTOR SHALL ARRANGE ALL SURVEY SETOUT TO BE CARRIED OUT BY A REGISTERED SURVEYOR PRIOR TO COMMENCEMENT OF WORKS.
7. THE CONTRACTOR TO VERIFY EXISTING PAVEMENT LEVELS AT SAWCUT LINES MATCH THE DOCUMENTED LEVELS. IF THE LEVELS ARE NOT CORRECT, CONTRACTOR IS TO ADVISE SUPERINTENDANT.
8. DO NOT OBTAIN DIMENSIONS BY SCALING DRAWINGS.
9. IN CASE OF DOUBT OR DISCREPANCY REFER TO SUPERINTENDENT FOR CLARIFICATION OR CONFIRMATION PRIOR TO THE COMMENCEMENT OF CONSTRUCTION.
10. WHERE NEW WORKS ABUT EXISTING THE CONTRACTOR SHALL ENSURE THAT A SMOOTH EVEN PROFILE, FREE FROM ABRUPT CHANGES IS OBTAINED. MAKE SMOOTH TRANSITION TO EXISTING FEATURES AND MAKE GOOD WHERE JOINED.
11. TRENCHES THROUGH EXISTING ROAD AND CONCRETE PAVEMENTS SHALL BE SAWCUT TO FULL DEPTH OF CONCRETE AND A MIN 50mm IN BITUMINOUS PAVING.
12. ALL CIVIL ENGINEERING DESIGN HAS BEEN DOCUMENTED UNDER THE ASSUMPTION THAT ALL NECESSARY SITE CONTAMINATION REMEDIATION WORKS HAVE BEEN SATISFACTORILY COMPLETED (IF APPLICABLE) AND THAT THE SITE IS NOT AFFECTED BY ANY SOIL STRATA OR GROUNDWATER TABLE CONTAMINATION.

CONSTRUCTION TOLERANCES

1. TRENCH FOUNDATIONS, UNDER CONCRETE SLABS AND LOAD BEARING ELEMENTS: +/- 25mm.
2, PAVEMENT SUBGRADES: CONFORM TO MRTS04 - GENERAL EARTHWORKS, CLAUSE 6 - GEOMETRICS. SUBGRADE CROSSFALL TO BE WITHIN 0.5% OF ABSOLUTE DOCUMENTED CROSSFALLS.
3. BATTERS: +/- 50mm AND NO STEEPER THAN THOSE SHOWN ON DRAWINGS AND ARE FULLY CONTAINED WITHIN THE SITE.
4. OTHER GROUND SURFACES: +/- 50mm PROVIDED THE AREA REMAINS FREE DRAINING AND MATCHES ADJACENT CONSTRUCTION WHERE REQUIRED.

SIGNAGE AND LINEMARKING

1. ALL SIGNAGE, LINEMARKING AND RRPM's ARE TO BE INSTALLED IN ACCORDANCE WITH THE CURRENT DEPT. OF TRANSPORT & MAIN ROADS 'MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES' (M.U.T.C.D.)
2. EXISTING LINEMARKING MADE REDUNDANT BY NEW LINEMARKING IS TO BE REMOVED BY ROUTING OR OTHER METHOD APPROVED BY DEPARTMENT OF TRANSPORT & MAIN ROADS.
3. ALL LINEMARKING TO BE REFLECTORISED WHITE. U.N.O LINEMARKING REGULARLY DRIVEN ACROSS BY TRAFFIC SHALL BE PLACED USING THERMOPLASTIC (I.E STOP, GIVE WAY, CONTINUITY LINES, ARROWS, ETC). PAVEMENT MARKING PAINT TYPE SHALL BE IN ACCORDANCE WITH : QLD-MUTCD-ROAD MARKING MATERIALS, WATERBORNE PAINT, FOR USE WITH DROP-ON BEADS.
4. ALL TRAFFIC SIGNAL PAVEMENT MARKINGS TO BE INSTALLED IN ACCORDANCE WITH THE CURRENT DEPT. OF TRANSPORT & MAIN ROADS 'MANUAL OF UNIFORM TRAFFIC CONTROL DEVICES' (M.U.T.C.D.)
5. THE NOSE OF ALL ISLANDS ARE TO BE PAINTED WITH WHITE REFLECTORISED PAINT.
6. ALL TRAFFIC SIGNS TO BE SIZE 'B' UNLESS NOTED OTHERWISE.
7. FOR GUIDE POST INSTALLATION & DETAILS, REFER QUEENSLAND TRANSPORT STANDARD DRAWING No.1356
8. FOR TRAFFIC SIGN SUPPORT DETAILS, REFER QUEENSLAND TRANSPORT STANDARD DRAWING No.'s 1360 & 1362.
9. ALL SIGN MATERIAL TO BE CLASS 1.
10. LINE MARKING SHALL BE SPOTTED OUT AND APPROVED PRIOR TO SPRAYING BY THE SUPERINTENDENT.
11. PARKING BAYS ADJACENT TO PATHWAYS ARE REQUIRED TO HAVE WHEEL STOPS. REFER TO AS2890.1 FOR ADDITIONAL DETAILS.

PAVEMENTS

1. PRIOR TO PLACING THE PAVING MATERIAL, THE SUBGRADE SHALL BE TRIMMED TO THE CORRECT LINE AND SHAPE AND SHALL BE COMPACTED TO A DENSITY NOT LESS THAN 98% OF THE STANDARD MAXIMUM DRY DENSITY.
2. THE CONTRACTOR SHALL CONFIRM THE DESIGN CBR WITH A MINIMUM OF 3 TESTS TAKEN AT SUBGRADE LEVEL. WHERE DISCREPANCY IS FOUND, CONTACT THE DESIGN ENGINEER.
3. PAVEMENT MATERIALS SHALL COMPLY WITH THE FOLLOWING DEPT. OF TRANSPORT AND MAIN ROADS SPECIFICATION MRTS05 UNBOUND PAVEMENTS:
   (a) BASE COURSE MATERIALS SHALL BE TYPE 2.1, GRADING B OR C.
   (b) SUBBASE COURSE MATERIALS SHALL BE TYPE 2.3, GRADING B OR C
   (c) SUBGRADE REPLACEMENT MATERIALS SHALL BE TYPE 2.5, GRADING B OR C
4. COMPACTION STANDARDS (U.N.O BY AUTHORITY CODES)

BASE	95% MMDD
SUBBASE	95% MMDD
SUBGRADE	98% SMDD
SUBGRADE REPLACEMENT/CBR15	98% SMDD

5. ALLOW THE FOLLOWING COMPACTION TESTING BY N.A.T.A. REGISTERED LABORATORY - NOT LESS THAN 1 TEST PER 25 LINEAL METRES OF 2 LANE ROAD PER PAVEMENT COURSE OR PART THEREOF OR 1 TEST PER 300m^2 PER PAVEMENT COURSE.
6. PAVEMENT MATERIAL SHALL BE COMPACTED IN LAYERS NOT LESS THAN 75mm AND NOT EXCEEDING 200mm COMPACTED THICKNESS.
7. BASE COURSE SHALL BE TRIMMED TO THE REQUIRED LEVELS IN ACCORDANCE WITH THE FINISHED SURFACE LEVELS AND THE REQUIRED MINIMUM SURFACING THICKNESS.
8. PAVEMENT MATERIALS SHALL BE EXTENDED UNDER KERBS, AND KERB & CHANNEL FOR A DISTANCE OF 150mm BEYOND THE BACK OF THE KERB AND SHALL BE COMPACTED AS FOR PAVEMENT MATERIALS. THE MINIMUM COMPACTED THICKNESS OF PAVEMENT MATERIAL UNDER KERBS AND KERB & CHANNEL SHALL BE 75mm.
9. MATCH NEW PAVEMENTS NEATLY AND FLUSH WITH EXISTING SURFACE.
10. PAVEMENT HOLD POINTS
    10.1. SUBGRADE PROOF ROLL AND PRIOR TO SET-UP AND FORM FOR CONCRETE POUR.
    10.2. PRESEAL PROOF ROLL
    10.3. INSPECTION OF FORMWORK / STEEL PRIOR TO CONCRETE POUR.
    10.4. SUBMISSION OF SUBGRADE AND BASE DENSITY TESTS.

ASPHALTIC CONCRETE

1. ALL ASPHALTIC CONCRETE (AC) WORK TO BE PREPARED AND CARRIED OUT IN ACCORDANCE WITH GOOD ASPHALTIC PAVING PRACTICE AS DESCRIBED IN AS2150-2005 "ASPHALT (HOT-MIXED) PAVING - GUIDE TO GOOD PRACTICE" AND CURRENT DEPT. OF TRANSPORT AND MAIN ROADS SPECIFICATION MRTS30.
2. ASPHALT SURFACING SHALL NOT COMMENCE UNTIL THE UNDERLYING PAVEMENT (EITHER NEW OR EXISTING) HAS BEEN INSPECTED AND APPROVED BY THE SUPERINTENDENT.
3. EXISTING SURFACES SHALL BE THOROUGHLY CLEANED, BY BROOMING OR OTHER APPROVED MEANS. ALL FOREIGN MATTER ADHERING TO THE PAVEMENT SHALL BE REMOVED BEFORE ANY TACK COATING OR ASPHALT SPREADING IS CARRIED OUT.
4. NEW PAVEMENT SHALL BE DAMPENED AND THEN THOROUGHLY BROOMED AND ANY FOREIGN MATERIAL REMAINING ON THE PAVEMENT AFTER BROOMING SHALL BE REMOVED BY OTHER MEANS BEFORE ANY APPLICATION OF PRIMER.
5. THE PRIMER USED SHALL SATISFY THE REQUIREMENTS OF AS2157. PRIMING SHALL BE CARRIED OUT AT LEAST 48 HOURS BEFORE ASPHALT IS APPLIED,
6. THE SELECTION OF THE TYPE, RATE AND TEMPERATURE OF APPLICATION OF THE PRIMER OR PRIMER SEAL SHALL TAKE INTO CONSIDERATION:
   (a) THE PERMEABILITY AND ABSORPTION CHARACTERISTICS OF THE PAVEMENT.
   (b) THE TYPE OF MATERIAL USED IN THE PAVEMENT
   (c) THE CONDITION OF THE PAVEMENT SURFACE
   (d) WHETHER OR NOT THE PAVEMENT WILL BE OPEN TO TRAFFIC AFTER PRIMING
   (e) THE PERIOD OF TIME BEFORE APPLICATION OF ASPHALT SURFACING
7. APPLICATION RATES FOR PRIMER SHALL BE:

TYPE OF SURFACE	PRIMER GRADE	APPLICATION RATE
LOW POROSITY (EXTREMELY HARD AND DENSE AFTER COMPACTION)	AMC00 OR AMC0	0.5 - 1.01 L/M^2
MEDIUM POROSITY (GRAVELS WITH SANDY SILT BINDERS)	AMC0 OR AMC1	0.5 - 1.01 L/M^2
HIGH POROSITY (WEAK FROM TEXTURED APPEARANCE)	AMC1 OR AMC2	0.8 - 1.41 L/M^2

8. PRIMING SHALL NOT BE CARRIED OUT IF THE PAVEMENT IS WET, RAIN THREATENS OR THE PAVEMENT OR AMBIENT TEMPERATURE IS LESS THAN 15°C.
9. TACK COAT SHALL BE SPRAYED ON THE EXISTING SURFACE FOR RESURFACING WORK. IT IS NOT REQUIRED ON A NEW PRIMER OR PRIMER SEAL UNLESS ORDERED BY THE SUPERINTENDENT.
10. APPLY THE TACK COAT LIGHTLY AND EVENLY WITH A FINE SPRAY OF APPROVED BITUMEN EMULSION AT A RATE EQUIVALENT TO 0.3 - 0.5 L/M^2. DILUTION WITH WATER MAY BE REQUIRED TO FACILITATE SPRAYING.
11. ANY OVERSPRAY OF THE TACK COAT OR PRIMER ONTO KERB AND CHANNEL, MEDIANS, GULLY PITS ETC. SHALL BE REMOVED.
12. ASPHALT SHALL BE PLACED UTILISING AN APPROVED SELF-PROPELLED MACHINE WITH AN EFFECTIVE SPREADING CAPACITY OF NOT LESS THAN 400 TONNES OF MIX PER 8 HOUR DAY OR OF A SUITABLE CAPACITY FOR WORK ON SMALL CONSTRUCTION AREAS AND FOOTPATHS. IT SHOULD INCLUDE THE FOLLOWING FEATURES:
   (a) MEANS OF PUSHING EACH MOTOR TRUCK DURING SPREADING
   (b) A RECEIVING HOPPER INTO WHICH MOTOR TRUCKS CAN DISCHARGE MATERIAL
   (c) DISTRIBUTING SCREWS TO PLACE THE MATERIAL EVENLY IN FRONT OF THE SCREED PLATE WITHOUT SEGREGATION
   (d) AUTOMATIC TAMPING OR VIBRATING DEVICE
   (e) AN ADJUSTABLE SCREED CAPABLE OF PROVIDING A SMOOTH EVEN SURFACE FREE FROM TEARS OR OTHER BLEMISHES TO A WIDTH NOT LESS THAN 3.5M FOR WORK ON ROADWAYS. PROVISION SHALL BE MADE FOR EASY ADJUSTMENT TO PERMIT LESSER WIDTHS
   (f) AN APPROVED SCREED HEATING DEVICE
   (g) EFFECTIVE STEERING SUCH THAT THE MIX CAN BE LAID TO A TRUE LINE
   (h) MEANS OF ADJUSTING DEPTH OF SPREAD BETWEEN 6MM AND 100MM (COMPACTED THICKNESS)
13. THE TEMPERATURE OF THE MIX WHEN IT IS TIPPED INTO A SPREADER SHALL NOT BE LESS THAN 130°C, NOR GREATER THAN 175°C. SPREADING SHALL COMMENCE AT A TEMPERATURE OF NOT LESS THAN 95°C NOR GREATER THAN 130°C. IT MAY BE NECESSARY TO COMPLETE COMPACTION AT HIGHER TEMPERATURES 95°C FOR THIN LAYERS (LESS THAN 50MM) IN COLDE AND WINDY CONDITIONS.
14. DO NOT HAND PLACE ASPHALT WITHOUT PRIOR APPROVAL FROM ENGINEER.
15. THE NUMBER OF JOINTS BOTH LONGITUDINAL AND TRANSVERSE SHALL BE KEPT TO A MINIMUM.
16. THE DENSITY AND SURFACE FINISH AT JOINTS SHALL BE SIMILAR TO THOSE OF THE REMAINDER OF THE LAYER.
17. COMPACTION SHALL BE CARRIED OUT USING SUCH EQUIPMENT AND TECHNIQUES AS ARE NECESSARY TO ACHIEVE A FIELD DENSITY OF AT LEAST 93% OF THAT OF A LABORATORY SPECIMEN OF THE SAME MIX, COMPACTED BY THE MARSHALL TEST METHOD.
18. THE FINISHED SURFACE OF ALL COMPACTED ASPHALT SHALL BE FINISHED TRUE TO GRADE AND PROFILE WITH SMOOTH JOINTS AND A NEAT FINISH AROUND MANHOLES AND OTHER ROAD SURFACE FITTINGS AND SHALL NOT VARY MORE THAN:
   (a) ±5MM FROM THE SPECIFIED PLAN LEVEL AT ANY POINT.
   (b) +5MM OR -0MM FROM THE BOTTOM OF A STRAIGHT EDGE.
   (c) +5MM OR -0MM ADJACENT TO OTHER ELEMENTS SUCH AS KERBS AND THE LIKE TO AVOID POOLING OF SURFACE WATER.
19. MEASURED CORES CUT FOR DENSITY TESTING SHALL HAVE 4 OUT OF 5 TESTS FALL WITHIN TOLERANCE. 1 OUT OF 5 TESTS MAY FALL LESS THAN 5MM OUTSIDE OF TOLERANCE
20. DO NOT STORE PLANT EQUIPMENT OR TRAFFIC NEWLY LAID ASPHALTIC CONCRETE PAVEMENTS WITHOUT PRIOR APPROVAL FROM THE ENGINEER.
21. DO NOT APPLY MARKING PAINTS UNTIL ASPHALT HAS CURED IN ACCORDANCE WITH PAINT MANUFACTURERS SPECIFICATIONS.

ENGINEERING CERTIFICATION

1. TO CERTIFY THE CONSTRUCTED CIVIL WORKS, A QUALIFIED EXPERIENCED ENGINEER IS TO VISIT THE SITE TO OBSERVE CONSTRUCTION TECHNIQUES AND VARIOUS ELEMENTS THAT MAY BE CONCEALED WHEN THE WORKS ARE COMPLETE. PLEASE NOTE THAT RPEQ CERTIFIED CONSTRUCTION CERTIFICATION CANNOT BE PROVIDED WHERE NORTHROP STAFF HAVE NOT ATTENDED INSPECTIONS AND ADEQUATE AS-CONSTRUCTED DOCUMENTATION HAS NOT BEEN SUPPLIED. CONTRACTOR IS LIABLE FOR ALL COSTS & RE-WORK REQUIRED WHERE ADEQUATE CONSTRUCTION INSPECTIONS/RECORDS HAVE NOT BEEN FACILITATED & SUPPLIED.
2. THIS SPECIFICATION ALLOWS FOR CERTIFICATION OF WORKS CONTROLLED BY A PRIVATE CERTIFIER FOR LAND DEVELOPMENT WORKS. THIS SPECIFICATION DOES NOT COVER CERTIFICATION REQUIREMENTS FOR AUTHORITIES SUCH AS THE LOCAL COUNCIL, DEPT. OF TRANSPORT AND MAIN ROADS OR SERVICE PROVIDER. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO DETERMINE AND PROVIDE ALL PROJECT SPECIFIC CONSTRUCTION COMPLIANCE (AS CONSTRUCTED) INFORMATION TO THE SATISFACTION OF THE STAKEHOLDER / AUTHORITY. DISCREPANCIES BETWEEN THIS SPECIFICATION AND SPECIFICATIONS OF OTHER EXTERNAL STAKEHOLDERS / AUTHORITIES IS TO BE REPORTED TO THE SUPERINTENDENT FOR CLARIFICATION.
3. THE CONTRACTOR IS TO AGREE WITH THE ENGINEER AN APPROPRIATE SITE VISIT SCHEDULE AND FEE ARRANGEMENT PRIOR TO COMMENCEMENT OF THE WORKS. THE CONTRACTOR SHALL ENSURE THAT THE ENGINEER CAN SAFELY ACCESS ALL CIVIL ELEMENTS TO BE REVIEWED. SITE VISITS ARE CONDUCTED DURING NORMAL BUSINESS HOURS. WE REQUIRE TWO (2) WORKING DAY NOTICE FOR ANY SITE VISIT.
4. TO PROVIDE CERTIFICATION THE ENGINEER MUST VISIT THE SITE TO OBSERVE.
   4.1. PAVEMENTS
      4.1.1. POOR SUBGRADE CONDITIONS
      4.1.2. PROOF ROLLING OF SUB-GRADE
      4.1.3. PROOF ROLLING OF PRE-SEAL
      4.1.4. PLACEMENT OF SUB-BASE COURSE, BASE COURSE AND WEARING COURSE.
      4.1.5. PLACEMENT OF STEEL REINFORCEMENT , DOWELS AND JOINT CRADLES PRIOR TO POURING OF CONCRETE
   4.2. EARTHWORKS
      4.2.1. TOPSOIL STRIP
      4.2.2. EARTHWORKS BATTER
      4.2.3. FILLING
   4.3. STORMWATER DRAINAGE
      4.3.1. DRAINAGE TRENCHES PRIOR TO BACKFILLING
      4.3.2. LEGAL POINT OF CONNECTION PRIOR TO BACKFILLING
      4.3.3. ANY OTHER DRAINAGE STRUCTURE THAT MAY BE CONCEALED DURING THE COURSE OF THE WORKS
   4.4. CONCRETE STRUCTURES
      4.4.1. PLACEMENT OF ANY STEEL REINFORCEMENT PRIOR TO CONSTRUCTION.
5. THE CONTRACTOR SHALL PROVIDE SURVEYED LEVELS, PREPARED BY A QUALIFIED SURVEYOR FOR SUBGRADE, SUB-BASE COURSE, BASE COURSE AND WEARING COURSE.
6. THE CONTRACTOR SHALL PROVIDE AS CONSTRUCTED DOCUMENTATION PREPARED BY A QUALIFIED PRACTISING SURVEYOR. THE AS CONSTRUCTED DRAWINGS SHALL CLEARLY SHOW, STORMWATER GRATE/ COVER LEVELS, STORMWATER PIT INVERT LEVELS AND CORRESPONDING INVERT LEVELS OF ANY INCOMING OR OUTGOING PIPES, DIAMETER OF ALL PIPES, DIMENSIONS AND VOLUME OF ON-SITE DETENTION FACILITIES, INVERT LEVELS OF ORIFICE PLATES, OVERFLOW WEIRS, BASE OF TANK FINISHED LEVELS OF PAVEMENTS. THE AS CONSTRUCTED DRAWINGS SHALL SHOW WHERE THE SIZE OR ALIGNMENT OF CIVIL ENGINEERING ELEMENTS WHEN THEY DEVIATE FROM THE DESIGN DOCUMENTATION.
7. THE AS CONSTRUCTED DRAWINGS SHALL BE STAMPED WITH THE FOLLOWING STATEMENT "THESE AS CONSTRUCTED DRAWINGS HAVE BEEN PREPARED BY [COMPANY NAME] AND ARE A TRUE AND ACCURATE REPRESENTATION OF THE CONSTRUCTED WORKS". EACH DRAWING SHALL BE SIGNED AND DATED BY THE SURVEYOR WHO PREPARED THE DRAWINGS.

THESE AS CONSTRUCTED DRAWINGS HAVE BEEN PREPARED BY [COMPANY NAME} AND ARE A TRUE AND ACCURATE REPRESENTATION OF THE CONSTRUCTED WORKS.

SIGNED... DATE..

NAME..

POSITION..

8. AS CONSTRUCTED DRAWINGS SHALL BE PROVIDED IN BOTH AUTOCAD AND PDF FORMAT.
9. CONTRACTOR IS TO UNDERTAKE A CCTV INSPECTION OF ALL STORMWATER DRAINAGE PIPELINES AND PROVIDE TO THE ENGINEER FOR APPROVAL.
10. THE CONTRACTOR SHALL PROVIDE ALL RELEVANT TEST CERTIFICATES PROGRESSIVELY THROUGHOUT THE DURATION OF THE WORKS. ALL TEST CERTIFICATES SHALL BE PREPARED BY A NATA REGISTERED LABORATORY. TEST CERTIFICATES ARE REQUIRED FOR SUBGRADE COMPACTION, COMPACTION OF PAVEMENT LAYERS, COMPACTION OF FILLING OPERATIONS, CONCRETE SLUMP TEST, AND CONCRETE STRENGTH TESTS. THE CONTRACTOR SHALL PROVIDE ALL RELEVANT VALIDATIONS BY A GEOTECHNICAL ENGINEER FOR ALL IMPORTED FILL
11. THE CONTRACTOR IS TO MAINTAIN COMPREHENSIVE CONSTRUCTION RECORDS (PHOTOS, AS-CONSTRUCTED SURVEY, TESTING RESULTS MATERIAL DOCKETS ETC) AND IS TO ISSUE TO NORTHROP AT THE END OF EACH WORKING WEEK, AS WELL AS A WELL ORGANISED AND COLLATED 'CONSTRUCTION COMPLIANCE PACKAGE' OF DATA AT THE COMPLETION OF WORKS.
12. EACH TEST CERTIFICATE WILL NOMINATE THE DATE AND TIME OF THE TEST AND PROVIDE A LOCATION OF WHERE THE TEST SAMPLE WAS TAKEN FROM.
13. THE CONTRACTOR SHALL ARRANGE FOR THE ENGINEER TO CONDUCT A FINAL VISIT TO REVIEW OF THE CONSTRUCTED WORKS. THIS WILL REVIEW WILL NOT TAKE PLACE UNTIL THE WAE DOCUMENTATION AND RELEVANT TEST CERTIFICATES HAVE BEEN RECEIVED.
14. IF DEFECTIVE OR INCOMPLETE WORK IS FOUND DURING THE FINAL INSPECTION ANOTHER INSPECTION MAY BE REQUIRED AT THE CONTRACTORS EXPENSE TO VERIFY THE RECTIFICATION WORKS HAVE BEEN COMPLETED.

BITUMEN SEALING

1. PAVEMENT PREPARATION
   1.1. THE SURFACE TO BE SEALED SHALL BE DRY AND BROOMED BEFORE COMMENCEMENT OF WORK TO ENSURE COMPLETE REMOVAL OF ALL SUPERFICIAL, FOREIGN OR LOOSE MATTER.
   1.2. IF APPROVED BY THE MANAGING CONTRACTOR, ALL DEPRESSIONS OR UNEVEN AREAS ARE TO BE TACK-COATED AND BROUGHT TO GENERAL LEVEL OF PAVEMENT WITH ASPHALT CONCRETE BEFORE SEALING COMMENCES.
2. MATERIALS
   2.1. BINDER SHALL BE CLASS 170 TO AS 2008 OR APPROVED PROPRIETARY MATERIAL FOR PRIMING AND PRIME SEALING.
   2.2. AGGREGATE SHAPE, DURABILITY AND WET TO DRY STRENGTH SHALL COMPLY TO AS2758 FOR CLASS 'N' AGGREGATES. A 20kg SAMPLE TO BE APPROVED BY THE MANAGING CONTRACTOR PRIOR TO USE.
   2.3. AGGREGATES SHALL BE DELIVERED UNIFORMLY PRECOATED, EXCESSIVE PRECOATING WILL RESULT IN AGGREGATES BEING REJECTED.
   2.4. FOR TWO COAT FLUSH SEALS, THE SIZE OF THE AGGREGATE FOR THE SECOND COAT, WHILE NORMALLY HALF THAT OF THE FIRST COAT, SHALL BE DIMENSIONALLY COMPATIBLE WITH THAT OF THE FIRST COAT.
   2.5. PRECOATING AGENTS SHALL BE COMPATIBLE WITH THE AGGREGATES AND BINDER TO BE USED.
3. DESIGN
   3.1. DESIGN OF SPRAYED BITUMINOUS SEALS SHALL BE CARRIED OUT IN ACCORDANCE WITH THE AUSTROADS (NAASRA) PUBLICATION "PRINCIPLES AND PRACTICE OF BITUMINOUS SURFACING, VOLUME 1 - SPRAYED WORK".
   3.2. WHERE NOT INDICATED ON THE DRAWINGS, PRIMES AND PRIMER SEALS SHALL BE DESIGNED TO REMAIN INTACT UNTIL FINAL SEALING TAKES PLACE, HAVING REGARD FOR THE TRAFFIC AND CLIMATIC CONDITIONS.
   3.3. UNLESS OTHERWISE SPECIFIED, BINDER APPLICATION RATES SHALL BE SELECTED TO FILL 85% OF THE THEORETICAL VOIDS OF THE MAT.
4. BITUMEN FLUSH SEALING
   4.1. BITUMEN FLUSH SEALS SHALL BE EITHER SINGLE OR DOUBLE COAT AS SHOWN ON THE DRAWINGS. eg 14/7 INDICATES A DOUBLE COAT FLUSH SEAL USING TWO APPLICATIONS OF BITUMEN AND AGGREGATE. THE FIRST AGGREGATE LAYER BEING OF 14mm NOMINAL SIZE, THE SECOND 7mm.
   4.2. COVER AGGREGATE SHALL BE SPREAD IMMEDIATELY AFTER SPRAYING OF BINDER. IN NO CASE SHALL SPREADING BE DELAYED MORE THAN 8 MINUTES,
5. RECORDS
   5.1. ALL SPRAY RECORDS AND AGGREGATE SUPPLY TONNAGE RECEIPTS SHALL BE RETAINED AND PASSED ON TO THE CONSULTING ENGINEER AS PART OF QUALITY ASSURANCE PROCEDURES.

CONCRETE SEALING

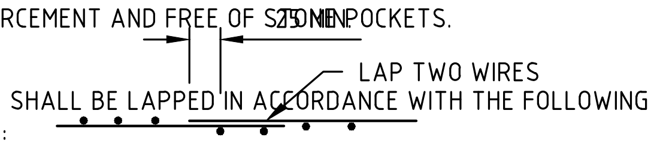
1. THE CURING PROCESS FOR NEW CONCRETE IS TO INCORPORATE THE FOLLOWING ASPECTS, GENERALLY AS ORDERED;
   1.1. SPRAY CURING COMPOUND
   1.2. SAWCUT JOINTS AS LOCATED AND SPECIFIED AS SOON AS CURING PERMITS.
   1.3. COVER NEW PAVING WITH HESSIAN AND BLACK PLASTIC SHEETS TAPED AT JOINTS ON COMPLETION OF SAWCUTTING. NOTE COVERING IS TO EXTEND A MINIMUM 5m BEYOND PAVEMENT BEING CURED. OVER ADJOINING (EXISTING) PAVEMENT AREAS, MAINTAIN CURING AS SPECIFIED.

CONCRETE PAVEMENTS

1. THIS SECTION REFERS TO CIVIL CONCRETE WORKS AND DOES NOT INCLUDE STRUCTURAL ELEMENTS SUCH AS BUILDINGS, BELOW GROUND STRUCTURES OR RETAINING WALLS.
2. ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS3600 CURRENT EDITION WITH AMENDMENTS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
3. CONCRETE QUALITY AND REINFORCING COVER
   ALL REQUIREMENTS OF THE CURRENT ACSE CONCRETE SPECIFICATION DOCUMENT 1 SHALL APPLY TO THE FORMWORK, REINFORCEMENT AND CONCRETE UNLESS NOTED OTHERWISE.

ELEMENT	CONCRETE STRENGTH f'c (MPa)	SPECIFIED SLUMP	NOMINAL AGGREGATE SIZE	MAX. 56 DAY DRYING SHRINKAGE	COVER (mm)
KERBS AND PATHS	25	60	20	650microns	TOP 40
PITS AND VEHICULAR PAVEMENTS	32	80	20	650microns	TOP 40

4. CONCRETE PROPERTIES SHALL BE VARIED FROM NORMAL CLASS AS FOLLOWS
   4.1. MINIMUM CEMENT CONTENT 250 kg/m^3
   4.2. MAXIMUM 56 DAY SHRINKAGE STRAIN = AS NOMINATED ABOVE
   4.3. PRIOR TO COMMENCEMENT CONCRETE SUPPLIER TO PROVIDE DRYING SHRINKAGE TEST RESULTS FROM PRODUCTION ASSESSMENT AS EVIDENCE THAT SPECIFIED DRYING SHRINKAGE LIMITS CAN BE ACHIEVED USING NORMAL MIX DESIGN.
5. ALL REINFORCEMENT SHALL BE FIRMLY SUPPORTED ON MILD STEEL PLASTIC TIPPED CHAIRS. PLASTIC CHAIRS OR CONCRETE CHAIRS AT NOT GREATER THAN 1m CENTRES BOTH WAYS. BARS SHALL BE TIED AT ALTERNATE INTERSECTIONS.
6. CEMENT TYPE SHALL BE (ACSE SPECIFICATION) TYPE SL.
7. PROJECT CONTROL TESTING SHALL BE CARRIED OUT IN ACCORDANCE WITH AS 1379. TEST CYLINDERS ARE TO BE KEPT ON SITE.
8. ALL COMPRESSIVE STRENGTH TEST REPORTS SHALL BE SUBMITTED TO THE CIVIL ENGINEER FOR REVIEW.
9. ALL CONCRETE IS TO BE CONTINUOUSLY CURED FOR A MINIMUM PERIOD OF 10 DAYS AFTER PLACING . CURING TO COMMENCE IMMEDIATELY AFTER FINISHING. SPRAY ON CURING COMPOUNDS TO COMPLY WITH AS3799.
10. PLACE CONCRETE CONTINUOUSLY BETWEEN CONSTRUCTION JOINTS SHOWN ON PLAN. DO NOT BREAK OR INTERRUPT SUCCESSIVE POURS SUCH THAT COLD JOINTS OCCUR. ANY REVISIONS OR ADDITIONS TO CONSTRUCTION JOINTS SHOWN ON PLAN REQUIRE APPROVAL FROM THE CIVIL ENGINEER.
11. FALLS IN SLAB AS SHOWN ON PLAN MAINTAIN MINIMUM SLAB THICKNESS AS SHOWN.
12. NO ADMIXTURES SHALL BE USED IN CONCRETE UNLESS APPROVED IN WRITING BY THE DESIGN ENGINEER.
13. THE FINISHED CONCRETE SHALL BE A DENSE HOMOGENOUS MASS, COMPLETELY FILLING THE FORMWORK, THOROUGHLY EMBEDDING THE REINFORCEMENT AND FREE OF STONE POCKETS.
14. FABRIC SHALL BE LAPPED IN ACCORDANCE WITH THE FOLLOWING DETAIL:

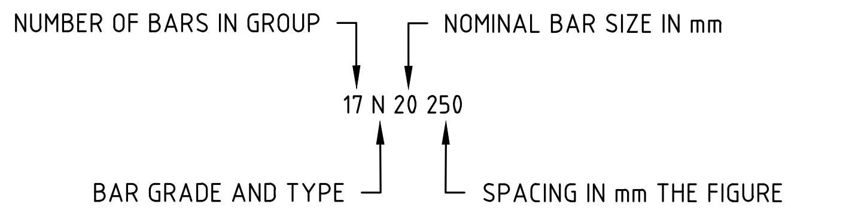


FOR FOLLOWING THE FABRIC SYMBOL SL IS THE REFERENCE NUMBER FABRIC TO AS1304.

15. POLYETHYLENE SHEET SHALL BE PLACED BELOW ALL CONCRETE PAVEMENTS.
16. ALL PENETRATIONS TO HAVE 2/N12 TRIMMER BARS TOP AND BOTTOM TO EACH FACE U.N.O. EXTEND TRIMMERS 700 BEYOND PENETRATION. MAINTAIN 40mm COVER TOP AND BOTTOM.
17. FORMWORK CLASS SHALL BE IN ACCORDANCE WITH AS3600.
18. SURFACE FINISHES:

ELEMENT	FORMWORK CLASS
STORMWATER PIT	OFF FORM
PAVEMENTS	MACHINE FLOAT / BROOM FINISH
KERBS	STEEL FLOAT / TROWEL

19. REINFORCEMENT SYMBOLS:
   N DENOTES GRADE 450 N BARS TO AS1302 GRADE N
   R DENOTES 230 R HOT ROLLED PLAIN BARS TO AS1302
   SL DENOTES HARD-DRAWN WIRE REINFORCING FABRIC TO AS1304



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DRN	AMENDMENTS	DATE	REV
JD	PRELIMINARY DESIGN 30%	12.09.25	A
JD	ISSUED FOR TENDER	23.12.25	B
JD	ISSUED FOR TENDER	28.01.26	C
LQ	ISSUED FOR TENDER	30.01.26	D
MM	ISSUED FOR APPROVAL	11.06.26	E

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Authorised For Issue S.KILLORAN 11.06.26	Construction Checked N/A

SKYGATE PLAY
BRISBANE AIRPORT

CIVIL SPECIFICATION NOTES
SHEET 2 OF 3

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Scale AS ABOVE
Size A1
Revision E

Drawing Number	Project No.	Originator	Volume	Level	Type	Discipline	Number
	25021	NRP	XX	XX	DR	CI	000011

File: 250946-C-000010-Spec.dwg
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